IXIGO TRAINS · PRODUCT CAPSTONE

Steps 1–8: research, problem, solution, PRD and AI extension

Best Way to Reach: turning a waitlisted ticket into a travel plan people trust

Rohit Kumar · Airtribe PM Launchpad · 6 November 2026

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Links

Live prototype: https://prototype-rohit-ixigo.lovable.app

Also in the submission: final deck (PDF), video walkthrough and Supporting Artifacts (PDF).

Part 1 – Step 1: Market & User Research

Oct 22, 2026 · @Rohit Kumar

Research goal and hypotheses

Goal: find out why only 26% of ixigo Trains users who hit a waitlisted or unavailable preferred train resolve a travel booking within 24 hours, and what would move them to a confident plan.

#	Hypothesis	How we tested it
H1	Users see alternatives but cannot compare them on what they care about (arrival time, station, certainty), so they keep re-searching instead of deciding.	Interviews (last-trip walkthrough); survey Q7, Q8, Q15
H2	Users do not trust a confirmation % on its own; they want to know "what happens to me if it does not confirm".	Interviews; survey Q13 trust score
H3	Alternatives are scattered across trains, buses and flights in separate tabs, so cross-mode comparison feels like too much work.	Interviews; survey Q7, Q15
H4	Price is not the main blocker; arrival time, station convenience and family needs often matter more.	Interviews; survey Q12 ranking
H5	Many users "park" a waitlisted ticket, then panic close to chart time and leak to Tatkal, buses or agents.	Interviews; survey Q6, Q9, Q10
H6	Users who resolve elsewhere do so because an agent, family member or another app gave them one clear answer.	Interviews with segments B and D; survey Q3, Q11
H7	A single "% confirm" blurs own berth (CNF), shared berth (RAC) and WL, so even a "correct" prediction feels wrong. Founder's own ticket at about 90% ended in RAC.	Interviews (expected vs actual outcome); survey Q14

Research questions

1. When your preferred train showed WL or no seats, what did you do in the next 10 minutes, and in the next 24 hours? (H1, H5)
2. Which factors decided your final choice: price, arrival time, journey time, station, class, confirmation chance, or cancellation refund? Rank your top three. (H4)
3. When an app shows a confirmation probability, how much do you trust it, and what would make you trust it more? (H2)
4. How did you compare other trains, dates, nearby stations, buses or flights, and what was hardest about that? (H1, H3)
5. If you booked a waitlisted ticket, what was your backup plan if it did not confirm? (H2, H5)

6. If you finally booked on another app, through an agent, or not at all, what made that option win? (H6)
7. If an app could hand you one ready plan for this trip, what would it need to show before you would book it? (H1, H2)
8. Has a high-chance ticket ever ended as RAC or stayed waitlisted? How did that change your trust in the prediction? (H7)

Research plan

Three methods ran between 22 Oct and 6 Nov 2026: interviews for depth, a bilingual survey for breadth, and desk research on five competitors.

Who we talked to: plan vs actual

Segment	Who	Planned	Actual
A	Active ixigo users	6	10
B	Check ixigo, book elsewhere	6	3
C	Using ixigo less	4	2
D	IRCTC or MakeMyTrip users	4	4
–	Unclear from audio	–	2
Total		20	21 records (about 20 people)

Methods: plan vs actual

Method	What it gives	Planned	Actual
Interviews (recorded calls, WhatsApp)	Real stories and decisions	20+	20 calls + 1 WhatsApp chat
Bilingual Google Form, 18 questions	Breadth, rankings, trust score	50+ responses	7 responses, 6 qualified
Competitor desk research	Features and gaps	4 apps	5 players + offline agents
Play Store review mining	Unprompted complaints	40–60 reviews	Not completed

Limits: the survey sample is small, so its numbers are directional; segments B and C are under-represented; several call transcripts were noisy, so quotes are paraphrased and each record carries an evidence-confidence label.

Survey questionnaire

A bilingual (English + Hindi) Google Form with 18 questions in 4 sections, about 4 minutes; anyone who had not hit a waitlist in the last 6 months skipped straight to submit.

#	Question	Type	Tests
1	Did your preferred train show WL, RAC or no seats in the last 6 months?	Yes / No screener	–

#	Question	Type	Tests
2	How often do you travel by train?	Single choice	–
3	Which apps or channels do you use to book?	Multi-select	H6
4	What was the trip for?	Single choice	–
5	How many people were travelling?	Single choice	H4
6	How many days before the journey were you booking?	Single choice	H5
7	What did you do after seeing WL or no seats?	Multi-select	H1, H3
8	How many times did you search before deciding?	Single choice	H1
9	What did you finally do?	Single choice	H5
10	Did you try Tatkal? What happened?	Single choice	H5
11	Where did you finally book?	Single choice	H6
12	Rank: price, arrival time, journey time, confirmation chance, station, comfort, free cancellation	Ranking grid	H4
13	How much do you trust an app's "% chance"?	Scale 1–5	H2
14	Has a 70%+ ticket ever ended differently than you expected?	Single choice	H7
15	What was the hardest part of finding an alternative?	Single choice	H1, H3
16	If an app showed one plan plus a backup, what must it show for you to book?	Open text	Solution input
17	Age group	Single choice	–
18	Can I call you for a 15-minute chat?	Optional contact	Recruitment

Interview script and user tracker

Each interview took 15–20 minutes over a recorded call or WhatsApp, in English or Hindi, and walked through the person's last trip where the preferred train was waitlisted or unavailable.

Script

1. Warm-up (2 min): How often do you travel by train? Which apps do you use to book?
2. Last trip story (5 min): Tell me about the last time your train showed WL or no seats. Route, date, who was travelling?
3. Next steps (4 min): What did you do right after? How many times did you search again? Did you check buses, flights, other dates?
4. Decision (4 min): What did you finally do, and on which app or with whom? What made that option win?
5. Trust (2 min): Did you see a confirmation % anywhere? Did you believe it? What happened in the end?

6. Ideal fix (2 min): If the app gave you one ready plan, what would it have to show before you booked it?

Approach: asked for real stories rather than opinions, asked "why" twice on the final choice, and did not pitch any solution.

User interaction tracker

ID	Segment	Channel	Route / context	What they said (paraphrased)	Insight	Hypothesis	Evidence
P01	Unclear	Call	ixigo + IRCTC	Waitlist uncertainty and what to do if it doesn't confirm	Needs clear WL status and fallback	H5	Low
P02	A	Call	Family, hotel, connecting trips	Values confirmation % and 3X Refund; books for family	Family and connecting journeys	H1, H2	Low
P03	Unclear	Call	ixigo vs IRCTC, Tatkal	Compared Tatkal vs regular, time vs cost	Trade-offs need to be visible	H4	Low
P04	A	Call	Father's trips	WL/RAC, possible 90% figure, general-ticket fallback	Clear WL/RAC status and alternatives	H5, H7	Low
P05	A	Call	Mumbai, festival season	Last-minute vs advance; festival alternatives	Festival peaks need alternatives	H1	Low
P06	A	Call	Varanasi, family, 2AC	Trusts ~85–90%; prefers 2AC comfort; would take flight or bus; unaware of 3X Refund	Comfort-led priority; refund discoverability	H2, H4	Good
P07	A	Call	Ahmedabad–Prayagraj	Books at ~80% chance; uses 3X Refund	Personal threshold	H2	Medium–Low
P08	A	Call	Mumbai–Delhi, 12–15 trips/yr	Books at 80%+, often gets RAC; wants short,	RAC surprise; punctualit	H4, H7	Good

ID	Segment	Channel	Route / context	What they said (paraphrased)	Insight	Hypothesis	Evidence
				on-time night trains; checks IRCTC for data mismatches	y priority; data trust		
P09	D	Call	Delhi–Prayagraj / Lucknow	Downloaded ixigo, never booked; prefers official IRCTC	Trust and habit block conversion	H6	Good
P10	D	Call	Delhi–Kanpur, MakeMyTrip	On-time arrival is top priority; books only at 80%+, else takes a car	Punctuality priority; threshold	H2, H4	Good
P11	D	Call	Mumbai–Patna, IRCTC	IRCTC feels easy; Tatkal and bus as alternatives	Established IRCTC habit	H6	Low
P12	B	Call	Patna–Delhi	Checks ixigo, books on IRCTC; worried about charges and voucher terms (may be P07)	Refund terms reduce trust	H2, H6	Low–Medium
P13	A	Call	Punjab–West Bengal	Books WL only at 80%+; else IRCTC / Premium Tatkal; avoids long flight layovers	Threshold; speed priority	H2, H4, H5	Good
P14	A	Call	Malda–Delhi, Durga Puja	~80% chance, 90% when urgent; plans ahead; Premium Tatkal	Threshold rises with urgency	H2	Good
P15	C	Call	ixigo, reduced use	Wants 90–95%; a high-chance ticket did not confirm; login friction cut	Prediction miss; login friction	H2, H7	Medium–Low

ID	Segment	Channel	Route / context	What they said (paraphrased)	Insight	Hypothesis	Evidence
				usage			
P16	A	Call	Bangalore–Kolkata, family	Values timings, availability, sorting; flight as alternative	Needs sorting by own priority	H1, H4	Medium–Low
P17	D	Call	IRCTC, general ticket	General ticket as last resort; bus, flight	Last-resort fallbacks	H5	Very low
P18	C	Call	Shiv Ganga Exp., night	ixigo feels more complicated than IRCTC	Simplicity for IRCTC users	H6	Low–Medium
P19	B	Call	Varanasi–Delhi	Uses ixigo for info, books on IRCTC (official)	Info-to-booking leak	H6	Low
P20	B	Call	Bangalore – Darbhanga	Checks ixigo, books on IRCTC	Info-to-booking leak	H6	Very low
P21	A	WhatsApp	Mumbai–Agra, Rajdhani, family	Booked WL at 95%, confirmed ~4 h before; books only above 90%; sometimes RAC; prefers flight over Tatkal; rates ixigo 4/5	Threshold; flight as fallback	H2, H5, H7	Good

Segments: A = active ixigo user, B = checks ixigo but books elsewhere, C = using ixigo less, D = IRCTC / MakeMyTrip user. Full synthesis: interview report (6 Nov 2026). Quotes are paraphrased because several audio transcripts are noisy.

Hypothesis check after 21 records (about 20 people)

#	Hypothesis	Verdict	Evidence
H1	Can't compare alternatives	Supported	8 named fallbacks (flight, bus, Tatkal, car) but weigh them differently; P16 asks for sorting
H2	Don't trust a bare %	Refined	8 use the % as a personal cut-off (80–95%), so they rely on it; trust breaks when it misses
H3	Options scattered across tabs	Partly	Fallbacks are found on IRCTC, flights or Tatkal separately; needs usability testing
H4	Price is not the main	Supported	6 named comfort, punctuality, speed, safety or

#	Hypothesis	Verdict	Evidence
	blocker		timing as top priority
H5	Late panic near chart time	Partly	Tatkal / Premium Tatkal and general ticket as last resorts (P11, P13, P14, P17)
H6	One clear recommendation wins elsewhere	New angle	7 book mainly on IRCTC or MakeMyTrip, citing official status, habit, fees or refund terms
H7	CNF vs RAC gap hurts trust	Supported	P08 often gets RAC at 80%+, P15 high-chance miss, P21 sometimes RAC, plus founder's 90% → RAC

New insights to carry into Steps 2–4: users have a personal threshold (show options above it first); 3X Refund terms need plain explanation before purchase; login and flow complexity push IRCTC-first users away.

Survey results

The bilingual survey got 7 responses; 6 had hit a waitlist in the last 6 months. The sample is small, so treat it as directional, but it matches the interviews: everyone searched repeatedly, trust in the % is low, and price matters less than time and certainty.

Finding	Result (n = 6)	Hypothesis
Searched 3+ times before deciding	6 of 6 (2 searched 10+)	H1
Hardest part: too many options or too many apps	4 of 6	H1, H3
Trust in an app's "% chance" (1–5)	Average 1.7; nobody above 3	H2
A 70%+ ticket ended differently	2 of the 4 who remember (1 RAC, 1 stayed WL)	H7
Tried Tatkal for this trip	5 of 6 (3 got confirmed)	H5
Final booking channel	IRCTC 3, ixigo 2, agent 1	H6
Final outcome	Confirmed train 4, flight 1, agent 1	–

What matters most (average rank, 1 = most important): journey time 2.3, confirmation chance 3.0, arrival time 3.3, class/comfort 4.2, station nearby 4.7, price 4.8, free cancellation 5.7. Price ranked sixth of seven, supporting H4.

Tension worth noting: interviewees use the % as a booking rule (8 of 20 have a cut-off), yet survey trust in it is 1.7/5. Users depend on a number they don't believe, which is why the Honest Outcome Meter explains it rather than just showing it.

What the app must show to book a plan + backup (Q16, open text): confirmed seat, reserved seat, price and comfort, a good train for family comfort.

Competitive analysis

Every major player now pairs a confirmation prediction with a paid 3X-refund protection, so protection alone is no longer a differentiator; none of them helps the user compare alternatives side by side and decide.

Player	How it helps when the train is waitlisted	Strength	Gap for the user
ixigo Trains	Waitlist prediction; Alternate Travel Plan pays 3X (flight/bus) or 2X (train) value if the ticket stays waitlisted; Assured for free cancellation	Strong prediction brand (ConfirmTkt) plus cross-mode inventory	Refund is mostly a coupon, only on selected trains; the user still has to find and book the alternative themselves
ConfirmTkt (ixigo group)	Alternates: same-train other-station quota, two-train splits, train + bus combinations	Finds hidden confirmed seats	Many raw options with no ranking by the user's priorities
IRCTC – VIKALP	Free opt-in; waitlisted passengers auto-shifted to an alternate train on the same route if seats are free	Free, official, no extra fare	No guarantee, no choice of train after opting in, outcome known only after chart
MakeMyTrip Trains	Trip Guarantee 3X value; same-train longer-route and nearby-station suggestions	Smart route hacks shown in search	Options are suggestions; trade-offs (arrival time, station distance) not explained
redRail by redBus	Confirmation probability; Alternate Trip paid add-on, 3X refund	Bus inventory built in	Add-on protects money, not the journey; does not raise confirmation chance
Airlines (direct)	Last-minute flights	Certain seat	Price spikes close to date; no link to the train search
Offline agents	Tatkal attempts, quota knowledge, one clear recommendation	Trust and a single answer	Extra cost, no transparency

Opportunity for ixigo: own the decision moment, a ranked, honest comparison of trains, dates, stations, buses and flights in one view, instead of only selling protection after the decision.

Travel decision personas

Four personas emerged from the 21 interview records; the first two are the main target for this project, and the other two shape trust and conversion work.

	Threshold Planner	Punctual Frequent Traveller	IRCTC Loyalist	Lapsed User
Who	Active ixigo user who books ahead, often for family	12–15 trips a year on one corridor	Checks ixigo or Paytm for information, books	Used ixigo for years, now books less

	Threshold Planner	Punctual Frequent Traveller	IRCTC Loyalist	Lapsed User
			on IRCTC	
Evidence	P06, P07, P13, P14, P21	P08, P10	P09, P11, P12, P19, P20	P15, P18
Decision rule	Books WL only above 80–95%	Short, on-time journey, ideally overnight	Books where it feels official	Wants 90–95% and an easy flow
Fallback	Flight rather than Tatkal; Premium Tatkal	Tatkal; car if below 80%	Tatkal, bus	Flight
What breaks	A high % ends in RAC; 3X Refund terms unclear	Often gets RAC; data mismatches with IRCTC	Fees, voucher terms, habit	Login friction; a high-chance ticket missed
What would help	Honest CNF / RAC / WL split; options above their cut-off shown first	Rank by arrival and on-time record	Clear fees and refund terms; official-status cues	Simple flow, honest odds, reliable login

Top barriers (validated by research)

The research confirms the primary barrier: low decision confidence. Users have options but can't compare them on their own priorities, and they lean on a percentage they don't trust.

1. **Can't compare options on their own priorities (primary).** All 6 survey respondents searched 3+ times; 4 of 6 found too many options or too many apps the hardest part; 6 of 20 interviewees named a different top priority. (H1, H3, H4)
2. **Rely on a % they don't trust (primary).** 8 of 20 interviewees book only above 80–95%, yet survey trust in the % averages 1.7 out of 5. (H2)
3. **RAC surprises.** 3 interviewees, and 2 of the 4 survey respondents who remember, saw a high-chance ticket end as RAC or stay WL. (H7)
4. **Habit and trust pull bookings to IRCTC (secondary).** 7 of 20 book mainly on IRCTC or MakeMyTrip; fees and 3X Refund terms add doubt. (H6)
5. **Friction for occasional users (secondary).** 2 found login or the flow harder than IRCTC.

Late panic near chart time (H5) came up less than expected: only 1 of 6 survey respondents said they found out too late, while 5 of 6 tried Tatkal as a fallback.

Sources

- ixigo Alternate Travel Plan terms
- ConfirmTkt Alternates – YourStory
- IRCTC VIKALP scheme – LatestLY
- MakeMyTrip train features – Travel Trends Today
- redRail Alternate Trip – GrabOn

Part 2 – Steps 2–8: Problem, Solution & PRD

Nov 6, 2026 · @Rohit Kumar

This doc covers Steps 2–8 of the ixigo Trains capstone: the problem and its cost, the solution and PRD, launch, metrics, risks and the AI extension. Evidence comes from 21 interview records (about 20 people) and a 6-response survey, detailed in the Step 1 research (Part 1 of this file).

Step 2 – Refined problem statement

When an ixigo Trains user's preferred train or class shows waitlisted or no seats, they get many alternatives but no trusted way to compare them, so they re-search, hesitate and leave. Only 26% reach a resolved booking within 24 hours; the goal is 42% in 6 months.

Who: travellers who already know their route and date, most often booking for festivals, family trips and urgent work travel, often in groups.

What they need: one clear answer to "how do I get there with the least risk?", with the trade-offs (price, arrival time, station, certainty) visible in one view and a backup if the first plan fails.

What breaks today:

- Alternatives are spread across separate searches for trains, dates, stations, buses and flights.
- A single "% chance to confirm" hides the real outcome; a 90% ticket can still end as RAC (shared berth), which users read as the app being wrong.
- Protection products (Alternate Travel Plan, 3X refund) protect money, not the journey, so users still don't know how they will actually travel.

User voice *(paraphrased from 21 interview records, about 20 people)*:

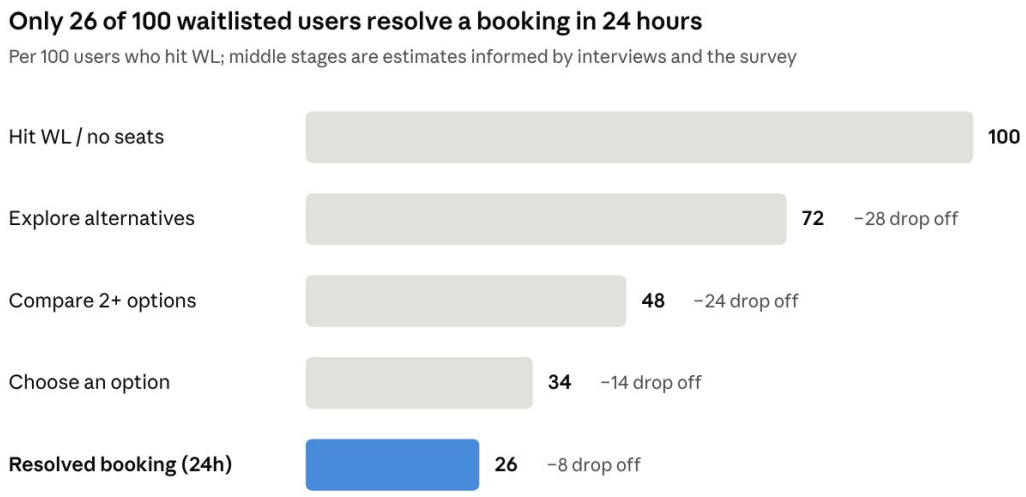
> *"I book a waitlisted ticket only if it shows 80% or more; otherwise I go by car." – P10, Delhi–Kanpur*
>
> *"I usually book at 80%+, and I often end up with RAC." – P08, Mumbai–Delhi, 12–15 trips a year*
>
> *"I book only above 90%. If there's no confirmed seat, I'd rather take a flight than fight for Tatkal." – P21, Mumbai–Agra*

Evidence: 8 of 20 use a personal cut-off of 80–95%; 3 were let down by high odds (RAC or no confirmation); 6 named a different top priority (comfort, punctuality, speed, safety, timing); 7 book mainly on IRCTC or MakeMyTrip.

Survey (6 qualified responses, directional): all 6 searched 3+ times before deciding; average trust in the "% chance" is 1.7 out of 5; 4 of 6 said too many options or too many apps was the hardest part; price ranked 6th of 7 behind journey time, confirmation and arrival time.

Founder observation: a ticket shown at ~90% chance to confirm ended as RAC; the trip happened, but trust in the prediction dropped.

Step 2 – Funnel



26% from the case brief; middle stages estimated from interviews and survey · per 100 users

The largest losses come between exploring and choosing (38 of 100), where users can't compare options or trust the odds; that is where both solutions act.

Step 2 – Primary barrier

The primary barrier is **low decision confidence**: users cannot compare alternatives on the trade-offs that matter to them, and they do not trust the certainty they are shown, so they keep searching instead of booking.

Why this barrier, not the others

Candidate barrier	Evidence	Fixable by ixigo?	Verdict
No inventory (no seats anywhere)	8 of 20 interviewees named working fallbacks (flight, bus, Tatkal, car); all 6 survey respondents eventually travelled	Out of scope	Not primary
Price of alternatives	Price ranked 6th of 7 in the survey, behind journey time, confirmation chance and arrival time	Partly; discounts are out of scope	Secondary
Can't compare + don't trust certainty	All 6 survey respondents searched 3+ times; 4 of 6 say too many options or apps; 8 of 20 book only above 80–95%, yet trust in the % averages 1.7/5; 3 saw high odds end in RAC or WL	**Yes, inside the app**	**Primary**
Found out too late	Only 1 of 6 survey respondents; Tatkal is the usual fallback (5 of 6 tried it)	Yes, via one timely alert	Secondary, folded into the solution
Habit and trust in IRCTC	7 of 20 book mainly on IRCTC or MakeMyTrip; fees and 3X Refund terms add doubt	Partly (clear fees and terms)	Secondary

Step 2 – Business impact

Moving resolution from 26% to 42% adds about 9.6 lakh resolved bookings a month and roughly ₹10 crore a month in net revenue at full run-rate, about ₹120 crore a year (assumptions below, to be tuned).

Assumptions *(not ixigo data; state them as estimates)*

- Users facing "no confirmed seat" each month: 60 lakh (from 104 M train passenger segments a year and high waitlist share on busy routes).
- Lift: 42% − 26% = 16 points → 9.6 lakh extra resolved bookings a month.
- Mode mix of extra bookings: 60% train, 30% bus, 10% flight.

Mode	Extra bookings / month	Avg ticket (₹)	Net revenue per booking (₹)	GMV / month (₹ cr)	Net revenue / month (₹ cr)
Train (alt train, class, date, protected WL)	5,76,000	1,200	60	69.1	3.5
Bus	2,88,000	1,100	110	31.7	3.2
Flight	96,000	6,500	350	62.4	3.4
Total	**9,60,000**			**163.2**	**10.0**

Other gains: fewer users lost to IRCTC, agents and other apps; higher repeat rate (already 85.55%) because users leave with a plan that worked; more cross-category users (train → bus/flight), which raises lifetime value.

Step 3 – Ideas and RICE prioritisation

Fourteen ideas were scored with RICE; the honest outcome meter, priority chips and the smart compare view lead because they reach every waitlisted user and attack the primary barrier directly.

RICE = Reach × Impact × Confidence ÷ Effort. Reach = lakh users a month who see it; Impact 0.25–3; Confidence %; Effort = person-months.

#	Idea	What it does	Reach (lakh/mo)	Impact	Confidence	Effort (PM)	RICE
1	Honest outcome meter	Shows CNF / RAC / WL split instead of one "% confirm"	60	2	80%	2	48.0
2	Priority chips	"Arrive by", "Cheapest", "Most certain", "Family together" re-rank options	60	1	70%	1	42.0
3	Smart compare view	One screen ranking trains, classes, dates, stations, bus, flight	60	3	80%	4	36.0
4	Chart-time decision nudge	"Decide by 6 PM tomorrow; after that flights rise" timeline	40	1	60%	1	24.0

#	Idea	What it does	Reach (lakh/mo)	Impact	Confidence	Effort (PM)	RICE
5	Explainable prediction	"WL 40 on this train cleared 7 of last 10 similar dates"	60	1	70%	2	21.0
6	AI trip resolver (TARA)	Chat that explains trade-offs and recommends a plan	60	2	50%	4	15.0
7	Group-fit filter	Shows only options with seats for all passengers together	20	2	70%	2	14.0
8	Post-trip regret check	One-tap "Was this the right choice?" to calibrate ranking	30	0.5	80%	1	12.0
9	Plan + backup bundle	Book WL and pre-select a bus or flight fallback with a clear trigger	30	3	60%	5	10.8
10	Share plan on WhatsApp	Send the plan to family for a quick yes	20	1	50%	1	10.0
11	Nearby station card	Shows a nearby station with cab time and cost	30	1	60%	2	9.0
12	Tatkal readiness	Opening-time reminder and pre-filled passengers	25	1	50%	2	6.3
13	Split-journey surfacing	Shows same-train longer route or two-train splits in compare	25	1	50%	3	4.2
14	Fallback flight price hold	Hold a flight fare for 24 hours while WL is pending	10	2	40%	4	2.0

Scores are estimates for prioritisation, not forecasts; reach uses the 60 lakh users a month from the impact model.

Step 3 – Shortlisted solutions and trade-offs

Two solutions go forward: **Best Way to Reach** (ideas 2 + 3 + 9-lite) fixes comparison, and **Honest Outcome Meter** (ideas 1 + 5) fixes trust; together they turn "many options" into "one plan I believe".

Solution A – Best Way to Reach

When the chosen train is WL or full, the user sees one ranked list of every alternative (other trains, classes, dates, nearby stations, bus, flight, protected WL) with a top pick, a backup, and priority chips to re-rank.

- Why: removes re-searching across tabs and apps; all 6 survey respondents searched 3+ times, and 4 of 6 called too many options or apps the hardest part.
- Priority chips because 6 of 20 interviewees named a different top priority (comfort, punctuality, speed, safety, timing), and the survey ranked journey time, confirmation and arrival above price.
- Reuses: ConfirmTkt Alternates, AbhiBus inventory, ixigo flights, Alternate Travel Plan.

Solution B – Honest Outcome Meter

Every waitlisted option shows the likely outcome as three bars, e.g. *Confirmed berth 55% · RAC 30% · Stays WL 15%*, plus one line of reason and "what happens if not".

- Why: 8 of 20 interviewees use the % as a booking rule (80–95%), yet survey trust in it averages 1.7 out of 5; 3 saw high odds end in RAC or stay WL.
- Users judge a prediction by the outcome they expected (own berth), so a 90% "confirm" that ends in RAC breaks trust even when the trip happens.
- Guardrail built in: never says "guaranteed"; ranges shown when data is thin.

Trade-offs

Decision	Option chosen	What we gain	What we give up
One ranked list vs separate tabs	One list	Faster decision, cross-mode bookings	Busier screen; needs careful design
Recommend a top pick vs neutral list	Top pick + backup	Users get a clear answer, like an agent	Risk of a wrong pick hurting trust; needs regret tracking
Show CNF / RAC / WL split vs one %	Split	Honest, builds long-term trust	Numbers can look less "positive"; may lower WL bookings short term
Build AI chat (idea 6) now vs later	Later (Step 8 demo only)	Ship faster, lower risk	Less "wow" in v1
Plan + backup bundle full build vs lite	Lite: suggest backup, user books it	1–2 months faster	Fewer automatic rescues in v1

Step 4 – User journey map

The biggest pain sits between "sees WL" and "decides": that is where Best Way to Reach and the Honest Outcome Meter act.

Stage	Today: user does	Thinks / feels	With the new experience
1. Search	Enters Mumbai → Patna, 5 Nov	Hopeful	Same
2. Sees WL	Preferred train shows WL 85, "40% chance"	Worried; "is 40% good?"	Outcome meter: CNF 25% · RAC 15% · WL 60%, one-line reason
3. Explores	Re-searches other trains, dates, then bus and flight apps	Overwhelmed, many tabs	"See best ways to reach" button opens one ranked list
4.	Notes prices and times on	Confused about	Priority chips (Arrive by / Cheapest /

Stage	Today: user does	Thinks / feels	With the new experience
Compares	paper or memory	trade-offs	Most certain / Family together) re-rank; top pick + backup
5. Decides	Calls family or agent, or gives up	Anxious, low trust	Books top pick in 2 taps, or protected WL with a named backup
6. Waits	Checks PNR many times	Stressed until chart	Chart-time nudge: "Your backup is ready if WL doesn't clear by 6 PM"
7. Travels / reflects	Feels the app was wrong if RAC	Trust drops	One-tap "Was this the right choice?" feeds ranking

Step 4 – PRD: Best Way to Reach + Honest Outcome Meter

Problem: waitlisted users can't compare alternatives or trust certainty, so only 26% resolve within 24 hours.

Goal: raise 24-hour resolved booking rate to 42% in 6 months without raising post-booking regret.

Users: ixigo Trains users with a known route and date whose preferred train or class has no confirmed seat. Priority personas: Threshold Planner and Punctual Frequent Traveller (Step 1 research doc).

In scope: alternate trains, classes, dates (±2 days), nearby stations (≤50 km), AbhiBus buses, ixigo flights, protected WL (Alternate Travel Plan / Assured).

Out of scope: inventory, IRCTC quota or Tatkal rules, pricing, refund policy, payments.

User stories

1. As a waitlisted traveller, I want all ways to reach my destination in one list so I don't search five times.
2. As a traveller with a deadline, I want to rank options by arrival time so I reach on time.
3. As a family traveller, I want only options with seats for all of us together.
4. As a cautious traveller, I want to see whether I'll get a berth, RAC or stay WL, so I'm not surprised.
5. As any traveller, I want a backup ready if my WL doesn't confirm.
6. As a traveller with a personal cut-off (say 80%), I want to see only options that meet it, so I don't waste time on long shots.
7. As a cautious buyer, I want protection terms in plain words before I pay, so I know what I get back and in what form.

Functional requirements

ID	Requirement	Priority	Evidence
FR1	When the selected train/class is WL or full, show a "See best ways to reach" entry on the result and on the booking page	P0	4 of 6 found too many options or apps the hardest part
FR2	One ranked list across trains, classes, dates, nearby stations, bus, flight, protected WL	P0	8 of 20 named fallbacks across modes
FR3	Each card shows price, depart/arrive time, duration, station distance, certainty, cancellation terms	P0	Journey time, confirmation, arrival ranked top 3

ID	Requirement	Priority	Evidence
FR4	Priority chips re-rank instantly; default = "Balanced"	P0	6 of 20 named different top priorities
FR5	"Top pick" and "Backup" labels with a one-line reason	P0	Users turn to agents and family for one clear answer
FR6	Outcome meter: CNF / RAC / WL % for every WL option; a range when data is thin	P0	3 saw high odds end in RAC or WL; founder's 90% → RAC
FR7	Never use "guaranteed" or "sure" for a prediction; show a "how we estimate" link	P0	Trust in the % is 1.7 out of 5
FR8	"My cut-off" filter: show waitlisted options only above the user's chosen chance (default 80%)	P0	8 of 20 book only above 80–95%
FR9	Group-fit toggle for 3+ passengers	P1	Family bookings common (P02, P04, P16, P21)
FR10	Chart-time decision nudge (max 1 push, user can switch off)	P1	Tatkal is the usual last resort (5 of 6)
FR11	Protection terms in plain words before purchase: what is refunded, cash vs voucher, expiry	P1	3X Refund unknown or unclear (P06, P12)
FR12	Post-trip "Was this the right choice?"	P1	Guardrail for wrong recommendations
FR13	"Updated X min ago" on availability and chances	P2	P08 checks IRCTC because of data mismatches
FR14	Share the plan on WhatsApp	P2	Families decide together (2 of 6 asked family or an agent)

Ranking logic (v1): score = weighted sum of certainty, arrival fit, price, journey time, station convenience; weights set by the chosen chip. Certainty for a confirmed seat = 1.0; for WL = P(CNF) + 0.5 × P(RAC).

Non-functional: list loads in under 2 seconds; works on 4G and low-end Android; Hindi and English copy.

Success metrics: see Step 6. **Launch plan:** see Step 5.

Open questions: latency of combining bus and flight inventory; whether RAC split data is available per train and quota.

Step 4 – Prototype and annotated screens

Clickable prototype: Best Way to Reach. Flow: search result → best ways to reach → option detail → plan set. Turn on "Show PM annotations" to see 10 numbered notes beside the screens.

Lovable prototype (submission link): prototype-rohit-ixigo.lovable.app. It includes the "My cut-off" filter (FR8), plain-language protection terms (FR11) and "updated 4 min ago" (FR13), with research evidence in each annotation.

Screen	Annotation	Design decision
1. Search result	Outcome meter	CNF / RAC / WL split replaces a single "% confirm"
2. Best ways to reach	Priority chips, top pick + reason, backup	One ranked list across train, bus, flight; reason shows the two factors that won
3. Option detail	"Last 10 similar Saturdays" history, backup toggle	Makes the odds concrete; backup turns protection into a plan
4. Plan set	Regret question	Guardrail and learning loop

Step 5 – Go-to-market

Launch inside the existing waitlist moment, not as a new product: the entry point appears only when a user hits WL or no seats, so there is no extra marketing cost.

Element	Plan
Target	Users whose preferred train/class shows WL or no seats on high-demand routes
Entry points	Search result card, booking page, PNR status page for existing WL tickets
Positioning	"See the best way to reach – with honest chances" / "पहुँचने का सबसे अच्छा तरीका देखें"
Channels	In-app only in v1; one opt-in chart-time push; no discounts or coupon blasts
Timing	Pilot before winter wedding season, scale before Holi (Mar) and summer vacation (May–Jun)
Partners	ConfirmTkt (prediction), AbhiBus (bus), ixigo flights; no new partners needed

Step 5 – Growth / behaviour loop

Each resolved trip makes the next one faster, because outcomes improve the ranking and the user learns to trust the app.

1. User hits WL and opens Best Way to Reach.
2. Gets a top pick + backup with honest odds, books in 2 taps.
3. Trip outcome (CNF, RAC, WL, bus, flight) and the regret answer are logged.
4. Prediction and ranking get better calibrated for that route.
5. User trusts ixigo more and comes back first next time, often sharing the plan with family on WhatsApp.
6. More users and outcomes → back to step 4.

Step 5 – Impact model by phase

Phase	Months	Users exposed	Resolution rate	Extra resolved bookings / mo	Net revenue / mo (₹ cr)
Baseline	0	60 lakh	26%	–	–

Phase	Months	Users exposed	Resolution rate	Extra resolved bookings / mo	Net revenue / mo (₹ cr)
Pilot	1–2	5% (3 lakh)	32%	18,000	0.2
Scale	3–4	50% (30 lakh)	37%	3,30,000	3.4
Full	5–6	100% (60 lakh)	42%	9,60,000	10.0

Revenue uses the Step 2 mix (60% train, 30% bus, 10% flight) at about ₹104 net revenue per extra booking.

Step 5 – Cohort rollout

Cohort	Who	Why first	Exit criteria to next cohort
1. Pilot	5% of WL users on 20 top routes (Mumbai–Patna, Delhi–Lucknow, etc.), A/B test	Highest pain, enough data	+4 pts resolution vs control; regret ≤ control
2. Families	Bookings with 3+ passengers, all routes	Biggest group-fit pain	Group-fit used by 30%+
3. All WL users	50% then 100%	Scale	Guardrails stable for 4 weeks
4. PNR-page users	People who already hold a WL ticket	Rescue before chart time	10%+ book a backup

Step 6 – Metrics

The North Star is the **24-hour resolved booking rate** for waitlisted users: 26% today, 42% in 6 months.

Definition: of users who hit WL or no seats for a route and date, the share who within 24 hours book a confirmed train, bus or flight for that journey, or knowingly book protected WL.

Supporting metrics

Metric	Type	Baseline → target
Best Way to Reach open rate among WL users	Leading	New → 60%
Searches before booking	Leading	3–5 for most (survey) → 1–2
Time from first WL to booking	Leading	−40%
Chip or cut-off filter used	Leading	New → 35%
Top pick or backup booked	Leading	New → 50% of bookings from the list
Cross-mode bookings (bus + flight share)	Lagging	+10 pts
Trust in the shown chance (in-app pulse, 1–5)	Lagging	1.7 (survey) → 3.5
90-day repeat booking rate	Lagging	85.6% → 88%

Metric	Type	Baseline → target

Guardrails

Guardrail	Why	Threshold
Post-booking regret ("wrong choice")	Bad recommendations	≤ 8%, not above control
Prediction calibration (predicted vs actual CNF, RAC, WL)	Overstated confidence	Gap ≤ 5 pts per bucket
Cancellation rate of recommended options	Users booked something that didn't fit	Not above control
Complaints mentioning "prediction wrong" or RAC	Trust	−30%
Page load time	Experience	< 2 s

Event tracking

Event	Trigger	Key properties
wl_seen	WL / no seats shown on search or booking	route, date, train, class, wl_no, pax, p_cnf, p_rac, p_wl
bwr_opened	User opens Best Way to Reach	entry_point, options_count
chip_selected	Priority chip tapped	chip, position_before, position_after
option_viewed	Card expanded	mode, rank, price, arrive_time, certainty
option_booked	Booking completed from the list	mode, rank, label (top_pick / backup / other), fare
backup_selected	Backup chosen for a WL ticket	backup_mode, trigger_time
chart_outcome	Chart prepared	final_status (CNF / RAC / WL), predicted split
regret_answered	Post-trip question answered	answer, mode, rank

Step 7 – Risks and mitigations

Risk	Likelihood	Impact	Mitigation
Wrong top pick (e.g. bus arrives late)	Medium	High	Regret tracking, conservative weights, always show a backup
Prediction overconfident, more RAC surprises	Medium	High	Show split + ranges, calibration guardrail, "how we estimate" link
Showing honest odds lowers WL bookings	Medium	Medium	Offer protected WL and backup on the same screen
Bus/flight inventory slow to	Medium	Medium	Load trains first, stream bus and flight cards

Risk	Likelihood	Impact	Mitigation
load			in
Users see it as upselling flights	Low	High	Default "Balanced" chip, cheapest option always visible
Notification fatigue	Low	Medium	Max one chart-time push, opt-out
Cannibalising train bookings with buses	Low	Low	Net revenue per bus booking is higher; overall resolution is the goal

Step 7 – Trade-offs

- Honesty over short-term conversion: a lower-looking % now buys trust and repeat use.
- Recommendation over neutrality: users want an answer like an agent gives, at the cost of owning wrong picks.
- Speed over completeness: v1 skips full plan-plus-backup automation and AI chat.

Step 7 – Constraints and assumptions

- **Constraints:** no change to railway inventory, IRCTC quota or Tatkal rules, pricing or refund policy; no reliance on discounts.
- **Assumptions:** 60 lakh WL users a month; ConfirmTkt can estimate RAC separately; AbhiBus and flight APIs return within 2 s; mode mix 60/30/10; interview findings from about 20 people generalise to the wider base.

Step 8 – AI trip resolver (vibe-coded)

Working artifact: Trip Resolver (screens in Part 3 of this file). The traveller types their situation in their own words (English or Hinglish); Claude turns it into priorities and limits; transparent code ranks the real options; the answer states chances honestly.

Where it fits the journey: it is the conversational front door to Best Way to Reach (journey stage 4, "Compares"). Instead of choosing chips, a user can say "we are 3, my mother is 65, need to reach by Sunday morning, budget ₹5,000" and get the same ranked list with the right weights. In ixigo this would live inside TARA / ixigo NEXT.

Logic

1. AI understands, it does not invent: the model returns only weights (certainty, arrival, price, time, station, together, comfort), hard limits (budget, modes to avoid, arrive-by, no sitting overnight), a summary and reply language. It never sees editable prices and never names a train.
2. Ranking is code: score = Σ weight × factor; options breaking a hard limit are removed and listed as "left out".
3. Backup = best confirmed option other than the top pick.
4. Honesty guardrail: answer shows own berth / RAC / stays WL; words like "guaranteed" are blocked.
5. Fallback: a keyword parser (English + Hinglish) answers when AI is unavailable.

Prompt (abridged)

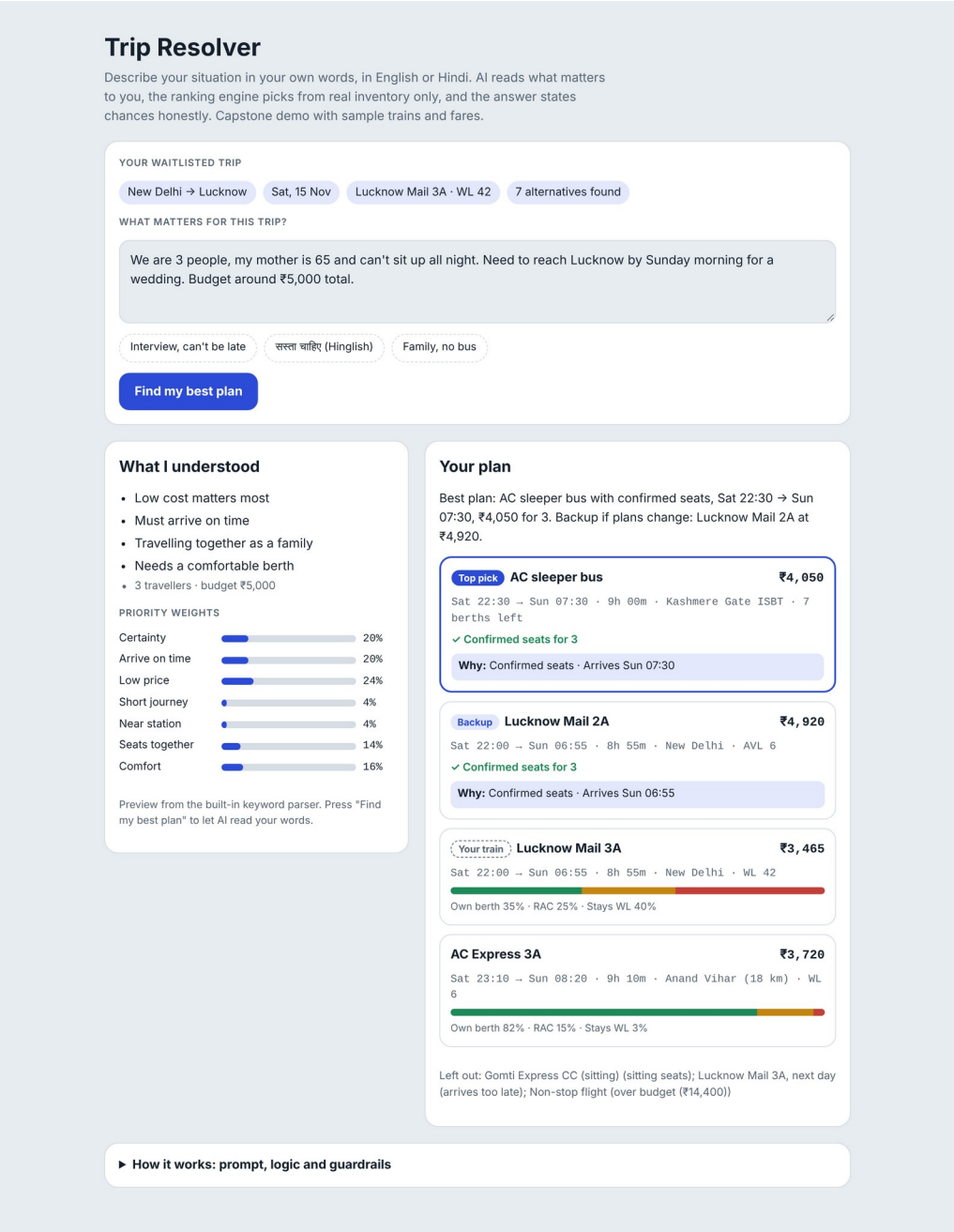
```
You help Indian train travellers whose preferred train is waitlisted. Read the
traveller's message and turn it into ranking preferences. Do not recommend a
specific train and do not invent options or prices.
```

```
Reply with only one JSON object:
{"understood": [3-5 bullets], "weights":
{"cert","arrive","price","time","station","together","comfort": 0-1},
"travellers", "budget_total_inr", "arrive_by", "avoid_modes",
"avoid_sitting_overnight", "reply_language"}
```

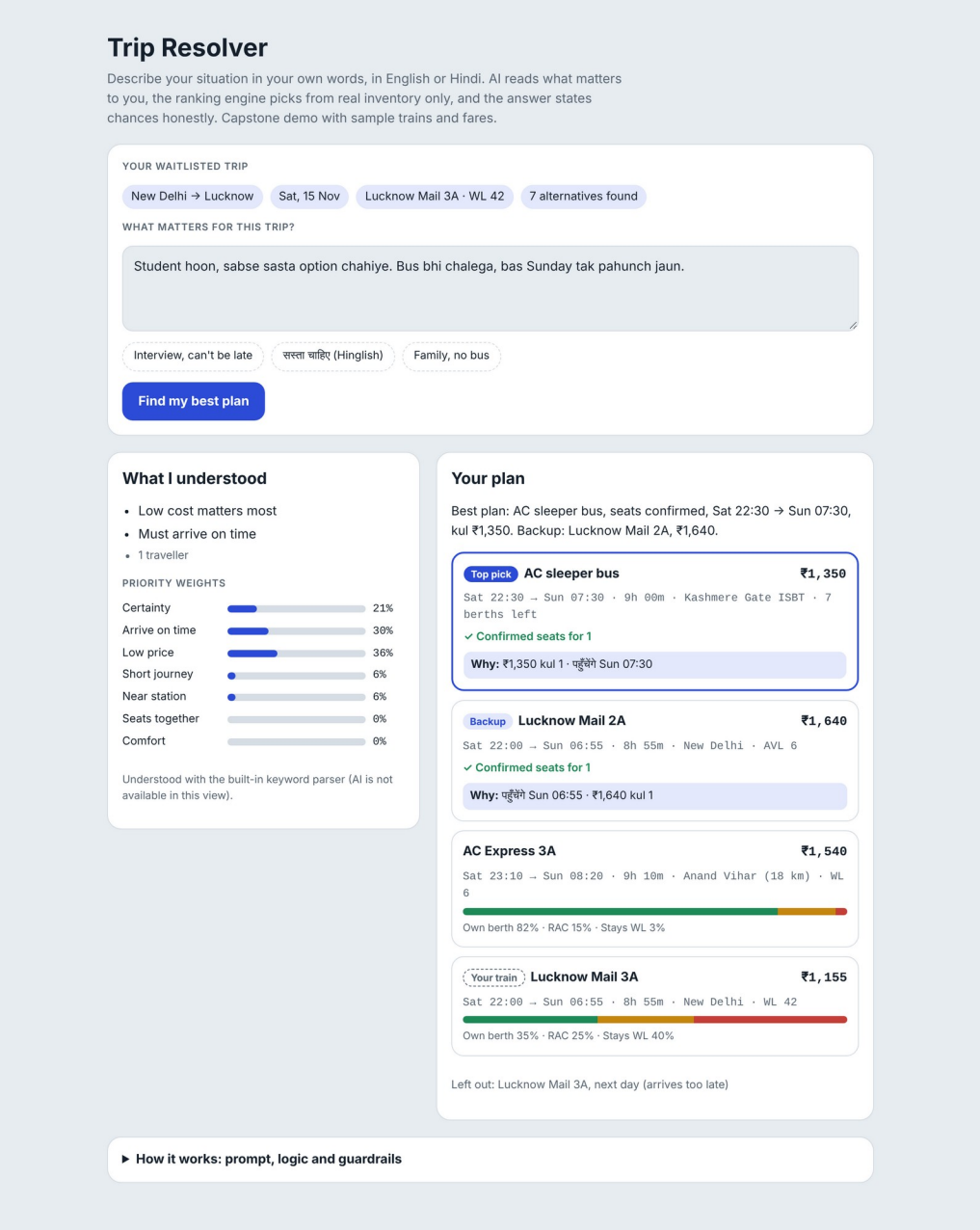
The full prompt is visible inside the artifact under "How it works".

Part 3 – Step 8: Trip Resolver screens

The Trip Resolver is the conversational front door to Best Way to Reach. The traveller describes the trip in their own words; the tool shows what it understood, the priority weights it applied, and a ranked plan with a top pick, a confirmed backup and honest odds. Sample trains and fares; the live AI run is shown in the video walkthrough.



Run 1 · “We are 3 people, my mother is 65 and can’t sit up all night. Need to reach Lucknow by Sunday morning for a wedding. Budget around ₹5,000 total.” Result: AC sleeper bus with confirmed seats as the top pick, Lucknow Mail 2A as backup; the user’s WL 42 train shows own berth 35% · RAC 25% · stays WL 40%; options breaking the limits (sitting overnight, over budget, arriving late) are listed as left out.



Run 2 (Hinglish) · “Student hoon, sabse sasta option chahiye. Bus bhi chalega, bas Sunday tak pahunch jaun.” Result: weights shift to low price, the cheapest confirmed option leads, and the reply mixes Hindi. This run used the built-in keyword parser, the fallback when AI is unavailable.

Other messages to try: “Solo trip for a job interview at 11 AM Sunday in Lucknow. I must not be late. Price doesn’t matter much.” · “Family of 3 with a small kid, we want to sit together and avoid buses. Fine with paying a bit more.”